DataCleanAI: Automated Data Quality, Cleaning & Validation Platform

Bonagani Prathusha
Department of Computer Science and Engineering, Koneru Lakshmaiah Education Foundation, Bowrampet, Hyderabad-500043, Telangana, India.
Email: prathushabonagani@gmail.com

Yakkala Guna Veera Sri Amith
Department of Computer Science and Engineering, Koneru Lakshmaiah Education Foundation, Bowrampet, Hyderabad-500043, Telangana, India.
Email: 2420030234@klh.edu.in

Kotte Manoj
Department of Computer Science and Engineering, Koneru Lakshmaiah Education Foundation, Bowrampet, Hyderabad-500043, Telangana, India.
Email: 2420030245@klh.edu.in

BVS Jayadeep
Department of Computer Science and Engineering, Koneru Lakshmaiah Education Foundation, Bowrampet, Hyderabad-500043, Telangana, India.
Email: 2420030381@klh.edu.in

***Abstract*—Problems with data quality like missing values, repeated records, extreme numeric values, and different text formats can decrease the reliability of reports and analyses to be done later on. The current paper describes DataCleanAI, a simple web platform that lets users check data and make them good, using one workflow. The developed prototype accepts and analyses CSV and Excel files, presents columns in profile form, shows missing values and identical records, visualizes numeric outliers based on the interquartile range, gives a score of data quality, and applies cleaning procedures. The cleaning processes include median for numeric types of information, mode to be used for categories, elimination of rows with empty data, deletion of same rows, elimination of outliers based on IQR, and normalization of words. A Streamlit interface presents statistics, reports on errors and comparisons. Working principles of the platform show users the transformations but not mysterious automatic repairs.**

***Index Terms*—data quality, data cleaning, data profiling, missing-value imputation, duplicate detection, outlier detection, Streamlit, data validation**

I. Introduction

Any application leveraging data analytics relies on both the effectiveness of analytical techniques employed and the quality of the information provided. Data from the real world may include missing values, duplicate entries, different naming conventions, non-logical figures, and dissimilar interpretations of the same category. Implemented faults may be transmuted into dashboards, simple statistics, and machine learning processes. Due to this reason, the process of data cleansing has been known to be of vital importance in data processing [1], [2]. Data quality is multi-dimensional. Completeness and uniqueness of data can be used, but data quality also consists of validity, consistency, relevance and context [3]. As a result of this some cleaning action can lead to improvement of some data aspect but will alter another at the same time. For example, replacing a missing number with the average value of the whole column allows data to be analyzed easier but primarily hides the fact that number in question was not measured. Similarly, an outlier may either be regarded as an anomaly or an exceptional observation.

DataCleanAI is designed as a prototype for a project that provides a web-based approach to carrying out various data-preparation tasks. A file such as CSV or Excel is first uploaded by the user to examine profile or summary of issues, and thereafter, appropriate cleaning actions can be selected before the results are exported. The focus of this system is in making the errors known and the changes made visible so that users may proceed with appropriate actions. The idea is not to promote the system as a new learning model or a substitute for domain-specific assessment approaches. Some of the contributions of the project may be outlined as follows: (1) unified interface integrating profiling, issue detection and cleaning; (2) a clear heuristic scores based on complete information, uniqueness, validity and consistency; (3) customizable approach to traditional cleaning techniques; and (4) differentiation in planned validation of functional assessment and additional verification of repairs made concerning the dataset.

II. Related Work

Rahm and Do have examined several instances of problems associated with data-cleansing processes, such as missing information and incompatible entries, as well as problems encountered during cooperation of different data sources [1]. Their study establishes a certain degree of background on the treatment of data cleaning as intelligence detection and transformation instead of data-cleansing as a single algorithm. Wang and Strong pointed out that the interpretation of quality in data should be made from the perspective of the data consumers, which goes beyond the notion of the data being accurate [3]. By being presented with this view, it becomes possible to see the rationale behind the present use of the multiple indicators in DataCleanAI. The score in the model only reflects a limited operational subset of quality indicators; it does not pretend to detect all the quality aspects. Interactive data preparation tools are looking into how visualization and discoveries as correction methods can facilitate analysis [4]. DataCleanAI applies the established theory according to which data transformation is something that ought to be made visible through the information.

Recent studies have explored the use of integrity constraints, external information, and statistical indicators to fix inconsistent data. Chu et al. provided an overview of the new techniques and challenges in automatic data cleansing [2], whereas Rekatsinas et al. have proposed HoloClean, a method based on the idea of holistic data cleansing [5]. These solutions have to tackle more complex data inference and cleansing situations than the prototype used in this study. However, in contrast to them, DataCleanAI emphasizes the importance of an understandable set of transformations that the user can choose from. The gap at the project level being addressed in this paper is practical integration: simple profiling, issues summary, the selected repairs, before-and-after comparison, and export are usually performed by different scripts or tools. DataCleanAI

brings all of these activities together in one simple interface, which is an innovative solution; however, no claim is made that the data-cleaning techniques used in this project are original. Table I summarizes the related studies and their relevance to DataCleanAI.

TABLE I
SUMMARY OF RELATED WORK

Study	Main focus	Relevance to DataCleanAI
Rahm and Do [1]	Data-cleaning problems and approaches	Motivates modular detection and repair stages
Chu et al. [2]	Survey/tutorial of modern data-cleaning challenges	Highlights rule-, pattern-, and statistics-based methods
Wang and Strong [3]	Consumer-oriented data-quality dimensions	Motivates a multi-component quality summary
Kandel et al. [4]	Interactive visual transformation scripts	Supports previewable, inspectable cleaning workflows
Rekatsinas et al. [5]	Probabilistic holistic data repair	Provides a more advanced comparison point for future work

III. Problem Statement and Objectives

The problem that is being resolved is the amount of work and uncertainty when preparing tabular data as the user must manually check different types of data errors and fulfill separate transformations. The desired initial platform should recognize typical errors, determine which operations should be used, keep the original version of data, and get the proper cleaned data. It should also not suggest that there exists one universal rule that can establish the real meaning of each 'suspicious' cell. The goals of DataCleanAI are:

- To input tabular data in CSV and Excel format through a user-friendly web interface.
- To give an overview of dataset dimensions, types, number of missing values, duplicated lines, and numeric outliers.
- To provide a meaningful heuristic quality score with indicators for each component.
- To allow the user to make decisions on imputation, lines deletion, duplicates deletion, outlier capping, and text-standardizing.
- To show the cleaned data and previous and future data to make transformations reviewable.
- To export the cleaned dataset in CSV format and allow for further verification and validation of the work done.

IV. Proposed System

DataCleanAI operates using multiple steps. First, the file supplied to the application is processed into a tabular format. Then, a profiler generates basic structure summaries and calculates missing-value records. Separate algorithms perform the search for exact duplicates in the dataset and possible anomalies in numeric columns. The score-related algorithm evaluates the required criteria that result in a final score between 0 and 100 calculated. At this stage, the user will mark the required options needed for cleansing, and the software will produce a data-cleaned output. The system will deliver a report on the performed tasks and make the cleansed data available for downloading. The core functional modules are:

- File processing and verification for CSV and .xlsx files.
- Data profiling and descriptive statistics.
- Reporting on missing values and exact duplicates.
- Detecting anomalies in numeric columns based on interquartile range.
- Heuristic assessments in regards to data quality.
- Adjustment of cleansing process and reporting with the visualization and comparison of results.

The system is designed for educational purposes and research work. Its functionality does not represent any area-related business rules; therefore, the reliability of detected anomalies and the appropriateness of missing data values cannot be guaranteed.

V. System Architecture and Data Flow

The prototype employs Python for data processing, along with Pandas and NumPy for table manipulation. Streamlit is used as the platform for the interface and Plotly for developing the interactive graphs. The model consists of two components including a presentation layer and reusable data processor functions. The interface takes care of file uploading, user operations and interactions, while the processing part of the prototype contains modules for data discovery, classification, cleaning and duplicate removal. The data flow of the idea consists of the following process:

- CSV/Excel uploaded → file reader → data frame in the memory.
- data frame → profiling system → numbers, summaries of columns and information about missing data.
- data frame → duplicating detection system and outliers detection → summary.
- profiling and summary data → the function for calculating quality score of data.
- initial data frame + options selected by user → cleaning process is done by the engine.
- raw report + result of cleaning process → presentation layer and visualization of data.

The uploaded original table is used as a baseline comparison in the current session, while the cleaning operation is based on its copy. The separation allows for a repetitive review within the session. The prototype is not defined as a permanent multi-user service, and some deployment-related aspects like authentication, storage of the database, and audit logs implementation have to be done explicitly.

VI. Data Science Methodology

A. Data Profiling: It calculates and provides data such as number of rows and columns regarding the columns missing values, data types and some obtained descriptive statistics. It serves as a descriptive measure as it determines the places where inspection is required but not whether its value is correct or not.

B. Handling of Missing Values: The user is given option to choose between the median/mode imputation and deleting the incomplete rows. In the case of numeric column median imputation is used. For non-numeric columns the mode (the most frequent value) is used. Although median imputation is more robust than mean imputation, it can still distort the relationship and distribution. Mode imputation may lead to

overestimation of the sizes of certain categories, while deleting incomplete records may result in losing useful information.

C. Duplicate Detection: Exact duplicate detection relies on row equality throughout all of its columns. It is useful for identifying repeated rows but is not qualified enough to detect near-duplicates, such as spelling variations or slightly different addresses. For these purposes record-linkage methods and similarity metrics are used.

D. Outlier Detection and Treatment: For each variable with numerical data, the prototype calculates Q1 - first quartile, Q3 - third quartile, and the interquartile range IQR = Q3 - Q1. The lower threshold is Q1 - 1.5 × IQR and the upper threshold is Q3 + 1.5 × IQR. The values falling below and above these thresholds are treated as outliers. If a capping option is chosen, then, the values are clipped. This is a general exploratory rule introduced by Tukey [6], it does not represent a wide acceptance of defining an outlier. It may not be appropriate for data with high skewness, small sample sizes, or other naturally heavy-tailed variables.

E. Standardising Text: For text-like columns, the prototype eliminates unnecessary leading and trailing spaces, merges repeated spaces into one space, and converts the text into title case. Although this can mitigate the differences in formatting somewhat, it can affect the way certain product codes, email, or case-sensitive identifiers look, thus it is advised to omit those columns from the process of standardisation in case the cases have a meaning.

F. Heuristic Quality Score: The four aspects calculated are completeness, uniqueness, validity and consistency. The total number of cells is represented by N, and the number of missing cells is represented by M. The formula for completeness is C = 100 × (1 − M/N). If D represents the ratio of duplicate rows to all rows, then the uniqueness score is calculated as U = 100 × (1 − D). The validity component V applies to a basic numeric-value check; in its current form, it is limited and should not be taken as a complete type of schema or domain validation.

Consistency is defined as S = (C + U)/2. The overall score is:

$$Q = 0.35C + 0.25U + 0.20V + 0.20S \quad (1)$$

The score is rounded and bounded to the 0–100 range. The weights are design choices, not weights estimated from expert labels or empirical outcomes. The score is therefore a comparative heuristic for the selected indicators and must not be described as a probability that the dataset is correct.

G. Report and Export on the Cleaning Process: The records contain information on the number of values completed, the number of skipped rows due to missingness, eliminated duplicates, modified numeric outlier values, and standardized text columns. The final table receives a clean row index and can be exported as CSV. The production-level system should also maintain reversible processing logs and verify output schema and conditions before export.

VII. Implementation

Using Streamlit, the project is being executed in Python language. Pandas provides reading of data frames, profiling, imputation, along with representation as well as processing of duplicates; NumPy helps in performing mathematical computation; Plotly is engaged in creating interactive charts. The code is modularized in terms of the functions for profiling, quality scoring, cleaning, duplicate detection, as well as outlier detection and file utilities. The user interface groups the high-level functions in terms of tabs: overview, profiling, issues, cleaning, visualizations, and export. The application accepts uploads of CSV and Excel files and provides a sample data for the client for demonstration. Also, the parameters for cleaning are chosen explicitly by the user and aren't applied automatically. The current version of the prototype works on regular tabular cleaning, and doesn't claim other capabilities such as trained anomaly detectors, fuzzy matching, automatic schema inference as well as the storage of the data quality rules. The stack of programming technology is summarized in Table II. Versions get to be documented when submitting for reproducibility proof while this paper doesn't claim package version reliability obtained through verification.

TABLE II
Technology Stack of the DataCleanAI Prototype

Component	Technology	Role
Interface	Streamlit	Upload controls, tabs, cleaning choices, metrics, and export
Data processing	Pandas	Data-frame ingestion, summaries, imputation, and transformations
Numerical operations	NumPy	Numeric column selection and related array operations
Visualization	Plotly	Interactive charts and before/after visual summaries
Testing	Python test modules	Unit checks for quality scoring and cleaning routines
Output	CSV export	Download of the cleaned data table

VIII. Results and Discussion

The present deliverable is both a working prototype and a base for code, not a completed benchmarking study. The process employed is set to generate four results: (1) a structural profile for the uploaded data; (2) issue summaries for missing cells, accurate duplicates, and IQR based outliers; (3) a heuristic score of quality with its indicators; and (4) a clean table with number of transformations. Table III provides a summary of the outputs and interpretational limits associated with these outputs. The flags only denote issues that need to be validated, not established truth. Similarly, a higher score after cleaning suggests the improvement of the chosen indicators but does not guarantee the correctness of repair or adequacy of the data for downstream operations.

TABLE III
Module Outputs and Interpretation Limits

Module	Input / operation	Output	Interpretation limit
Profiling	Rows, columns, types, missingness	Dataset summary and column profile	Descriptive, not a correctness audit
Duplicate analysis	Exact row equality	Duplicate row count and details	Near-duplicates are not detected
Outlier analysis	Numeric values and IQR bounds	Potential outlier counts by column	Rare valid values may be flagged

Module	Input / operation	Output	Interpretation limit
Quality scoring	Completeness, uniqueness, validity, consistency	Heuristic score from 0 to 100	Not calibrated against expert labels
Cleaning	User-selected operations	Cleaned data and action counts	Repairs need contextual review
Export	Cleaned data frame	CSV file	Schema and business rules need validation

No numerical benchmark results are reported here because a controlled experiment with a defined dataset, known injected errors, baseline methods, and repeated measurements has not been completed. Reporting an invented accuracy percentage, processing-time reduction, or superiority claim would be misleading. The next evaluation should use both synthetic datasets with controlled defects and public real-world datasets, recording dataset size, defect types, runtime, and repair correctness.

A suitable evaluation would compare the original and cleaned tables using precision and recall for defect detection, precision of repairs against known ground truth, number of altered cells, runtime, and memory consumption. For user-facing utility, a small usability study could record task completion time and user-rated clarity. Quality-score changes should be reported separately from repair precision, since the score is computed from the same indicators the cleaning operations are intended to change.

IX. Limitations

- The quality score uses manually chosen weights and a limited set of indicators; it is not empirically calibrated.
- The current validity calculation is a basic heuristic and does not check arbitrary schemas, allowed ranges, foreign keys, or domain constraints.
- Exact duplicate detection does not resolve fuzzy or entity-level duplicates.
- Median/mode imputation may bias distributions and relationships; row deletion may cause data loss.
- IQR-based outlier capping can suppress meaningful extreme observations and is not appropriate for every distribution.
- Title-case normalization can damage acronyms, identifiers, and values for which case is meaningful.
- The prototype does not yet provide comprehensive audit trails, role-based access, database persistence, or enterprise privacy controls.
- No controlled empirical evaluation has yet established detection precision, repair accuracy, scalability, or user benefits.

X. Future Scope

Future development should prioritize a reproducible evaluation suite with injected missingness, duplicates, text variations, invalid values, and known numeric anomalies. Each issue type should have a labeled reference set so detection precision, recall, and repair correctness can be measured. The study should compare the current rule-based approach with simple baselines and report runtime and memory use across increasing dataset sizes.

Additional capabilities could include configurable schema rules, range and type validation, date parsing, category standardization dictionaries, fuzzy duplicate matching, Isolation Forest or other anomaly-detection methods, and support for reusable cleaning recipes. Advanced repair approaches could combine integrity constraints and statistical evidence, as explored by systems such as HoloClean [5]. Such extensions should be optional and should expose confidence or review requirements rather than silently changing data.

Operational improvements include a reversible transformation history, downloadable cleaning reports, reproducible configuration files, larger-file handling, stronger file validation, accessibility improvements, and privacy-conscious deployment. Any machine-learning extension should be assessed on representative datasets and should not be presented as reliable merely because it changes an anomaly score or increases the platform's heuristic quality score.

XI. Conclusion

This paper presented DataCleanAI, a lightweight web-based platform for automated assistance with tabular data profiling, quality assessment, cleaning, visualization, and export. The prototype combines missing-value handling, exact duplicate removal, IQR-based outlier screening and capping, text normalization, and a transparent multi-component quality score within a single workflow. Its main project contribution is the integration of common preparation steps with visible controls and a cleaning report, rather than the invention of a new cleaning algorithm.

The current implementation is best treated as a demonstrator and a foundation for systematic evaluation. The quality score is heuristic, some cleaning operations can alter legitimate information, and no empirical performance claims are made in the absence of a controlled benchmark. Future work should establish ground-truth test data, measure defect detection and repair quality, expand domain-aware validation, and add auditable, reversible transformations. These steps are necessary before the platform can be recommended for high-stakes or production data workflows.

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